

axiomTM



The 30 Year Horizon

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Volume 3: Axiom Programmers Guide

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New Foreword

On October 1, 2001 Axiom was withdrawn from the market and ended life as a commercial product. On September 3, 2002 Axiom was released under the Modified BSD license, including this document. On August 27, 2003 Axiom was released as free and open source software available for download from the Free Software Foundation's website, Savannah.

Work on Axiom has had the generous support of the Center for Algorithms and Interactive Scientific Computation (CAISS) at City College of New York. Special thanks go to Dr. Gilbert Baumslag for his support of the long term goal.

The online version of this documentation is roughly 1000 pages. In order to make printed versions we've broken it up into three volumes. The first volume is tutorial in nature. The second volume is for programmers. The third volume is reference material. We've also added a fourth volume for developers. All of these changes represent an experiment in print-on-demand delivery of documentation. Time will tell whether the experiment succeeded.

Axiom has been in existence for over thirty years. It is estimated to contain about three hundred man-years of research and has, as of September 3, 2003, 143 people listed in the credits. All of these people have contributed directly or indirectly to making Axiom available. Axiom is being passed to the next generation. I'm looking forward to future milestones.

With that in mind I've introduced the theme of the "30 year horizon". We must invent the tools that support the Computational Mathematician working 30 years from now. How will research be done when every bit of mathematical knowledge is online and instantly available? What happens when we scale Axiom by a factor of 100, giving us 1.1 million domains? How can we integrate theory with code? How will we integrate theorems and proofs of the mathematics with space-time complexity proofs and running code? What visualization tools are needed? How do we support the conceptual structures and semantics of mathematics in effective ways? How do we support results from the sciences? How do we teach the next generation to be effective Computational Mathematicians?

The "30 year horizon" is much nearer than it appears.

Tim Daly
CAISS, City College of New York
November 10, 2003 ((iHy))

Chapter 1

Details for Programmers

Axiom maintains internal representations for domains. There are functions for examining the internals of objects of a particular domain.

1.1 Examining Internals

One useful function is **devaluate** which takes an object and returns a Lisp pair. The CAR of the pair is the Axiom type. The CDR of the pair is the object representation. For instances, consider the session where we create a list of objects using the domain **List(Any)**.

```
(1) -> w:=[1,7.2,"luanne",3*x^2+5,_,
          (3*x^2+5)::FRAC(POLY(INT)),_,
          (3*x^2+5)::POLY(FRAC(INT)),_,
          (3*x^2+5)::EXPR(INT)]$LIST(ANY)

          2      2      2      2
(1)  [1,7.2,"luanne",3x  + 5,3x  + 5,3x  + 5,3x  + 5]
                                           Type: List(Any)
```

The first object, **1** is a primitive object that has the domain **PI** and uses the underlying Lisp representation for the number.

```
(2) -> devaluate(1)$Lisp

(2)  1
                                           Type: SExpression
```

The second object, **7.2** is a primitive object that has the domain **FLOAT** and uses the underlying Lisp representation for the number, in this case, itself a pair whose CAR is the floating point base and whose CDR is the mantissa,

```
(3) -> devaluate(7.2)$Lisp
```

```
(3) (265633114661417543270 . - 65)
```

Type: SExpression

The third object, "luanne" is from the domain **STRING** and uses the Lisp string representation.

```
(4) -> devaluate("luanne")$Lisp
```

```
(4) luanne
```

Type: SExpression

Now we get more complicated. We illustrate various ways to store the formula $3x^2 + 5$ in different domains. Each domain has a chosen representation.

```
(5) -> devaluate(3*x^2+5)$Lisp
```

```
(5) (1 x (2 0 . 3) (0 0 . 5))
```

Type: SExpression

The fourth object, $3x^2 + 5$ is from the domain **POLY(INT)**. It is stored as the list

```
(1 x (2 0 . 3) (0 0 . 5))
```

From the domain **POLY** (Vol 10.3, POLY) we see that

```
Polynomial(R:Ring): ...
== SparseMultivariatePolynomial(R, Symbol) add ...
```

So objects from this domain are represented as **SMP(INT,SYMBOL)**. From this domain we see that

```
SparseMultivariatePolynomial(R: Ring, VarSet: OrderedSet): ...
== add
--representations
D := SparseUnivariatePolynomial(%)
```

So objects from this domain are represented as a **SUP(INT)**

```
SparseUnivariatePolynomial(R:Ring): ...
== PolynomialRing(R, NonNegativeInteger) add
```

So objects from this domain are represented as **PR(INT,NNI)**

```
PolynomialRing(R:Ring, E:OrderedAbelianMonoid): ...
FreeModule(R, E) add
--representations
Term:= Record(k:E, c:R)
Rep:= List Term
```

So objects from this domain are represented as **FM(INT,NNI)**

```
FreeModule(R:Ring,S:OrderedSet):
  == IndexedDirectProductAbelianGroup(R,S) add
  --representations
  Term:= Record(k:S,c:R)
  Rep:= List Term
```

So objects from this domain are represented as **IDPAG(INT,NNI)**

```
IndexedDirectProductAbelianGroup(A:AbelianGroup,S:OrderedSet):
  == IndexedDirectProductAbelianMonoid(A,S) add
```

So objects from this domain are represented as **IDPAM(INT,NNI)**

```
IndexedDirectProductAbelianMonoid(A:AbelianMonoid,S:OrderedSet):
  == IndexedDirectProductObject(A,S) add
  --representations
  Term:= Record(k:S,c:A)
  Rep:= List Term
```

So objects from this domain are represented as **IDPO(INT,NNI)**

```
IndexedDirectProductObject(A:SetCategory,S:OrderedSet):
  == add
  -- representations
  Term:= Record(k:S,c:A)
  Rep:= List Term
```

```
(6) -> devaluate((3*x^2+5)::FRAC(POLY(INT)))$Lisp
```

```
(6) ((1 x (2 0 . 3) (0 0 . 5)) 0 . 1)
```

Type: SExpression

```
(7) -> devaluate((3*x^2+5)::POLY(FRAC(INT)))$Lisp
```

```
(7) (1 x (2 0 3 . 1) (0 0 5 . 1))
```

Type: SExpression

```
(8) -> devaluate((3*x^2+5)::EXPR(INT))$Lisp
```

```
(8) ((1 [[x,0,%symbol()()()],NIL,1,1024] (2 0 . 3) (0 0 . 5)) 0 . 1)
```

Type: SExpression

```
(9) -> devaluate(w)$Lisp
```

```
(9)
```

```
((PositiveInteger) . 1) ((Float) 265633114661417543270 . - 65)
```

```

((String) . luanne) ((Polynomial (Integer)) 1 x (2 0 . 3) (0 0 . 5))
((Fraction (Polynomial (Integer))) (1 x (2 0 . 3) (0 0 . 5)) 0 . 1)
((Polynomial (Fraction (Integer))) 1 x (2 0 3 . 1) (0 0 5 . 1))

((Expression (Integer))
 (1 [[x,0,%symbol()()()],NIL,1,1024] (2 0 . 3) (0 0 . 5)) 0 . 1)
)

```

Type: SExpression

1.2 Makefile

This book is actually a literate program[Knut92] and can contain executable source code. In particular, the Makefile for this book is part of the source of the book and is included below.

Bibliography

- [Knut92] Donald E. Knuth. *Literate Programming*. Center for the Study of Language and Information, Stanford CA, 1992, 0-937073-81-4.

